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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:30:03",
  "variant" : 1,
  "subjectName" : "operating-systems",
  "categories" : [ "Process Scheduling", "Memory Management", "File Systems" ],
  "questions" : [ {
    "question" : "What state does a process enter if it is waiting for an I/O operation to finish?",
    "answers" : [ "Blocked / Waiting state", "Zombie state", "Ready state", "Running state" ],
    "correctIndex" : 0
  }, {
    "question" : "What is a major disadvantage of the Shortest Job First scheduling algorithm?",
    "answers" : [ "Massive memory cache utilization footprints", "Long-running processes can suffer from starvation", "It requires specialized graphical hardware layers", "High context-switching processing overhead" ],
    "correctIndex" : 1
  }, {
    "question" : "What is the main difference between internal and external fragmentation?",
    "answers" : [ "Internal occurs inside fixed allocated blocks; external occurs between variable blocks", "External happens inside kernel space; internal happens in user space", "External happens on hard drives; internal happens inside CPU registers", "Internal is fixed by defragmentation; external requires virtual memory flushes" ],
    "correctIndex" : 0
  }, {
    "question" : "What is the function of the Memory Management Unit (MMU)?",
    "answers" : [ "To allocate network sockets to background daemons", "To clear cache registers when a context switch occurs", "To map virtual addresses generated by the CPU to physical memory locations", "To partition hard drive sectors dynamically at runtime" ],
    "correctIndex" : 2
  }, {
    "question" : "What is a page fault?",
    "answers" : [ "A crash caused by syntax compile errors", "A physical hardware failure inside RAM modules", "An interrupt triggered when a program accesses a page not currently mapped in RAM", "A data anomaly where two files share the same sector" ],
    "correctIndex" : 2
  }, {
    "question" : "What state does a system enter when it spends more time swapping pages than executing instructions?",
    "answers" : [ "Thrashing", "Paging", "Fragmenting", "Starvation" ],
    "correctIndex" : 0
  }, {
    "question" : "What does an inode store in a Unix-like file system?",

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"answers" : [ "The literal plain-text contents of the target file", "File metadata, size, permissions,
and block pointers, but not the file name", "The human-readable file name and directory path
string", "The encryption keys used for secure network transmissions" ],
"correctIndex" : 1
}, {
"question" : "What structural pattern does a Hard Link follow compared to a Symbolic Link?",
"answers" : [ "Hard links require administrative privileges to execute", "Symbolic links can only
reference binary files; hard links handle folders", "A Hard Link points directly to the same inode; a
Symbolic Link points to a path string", "Hard links delete the original file target when broken;
symbolic links do not" ],
"correctIndex" : 2
}, {
"question" : "What component maps file paths to matching inode numbers in a file system
directory?",
"answers" : [ "File allocation tables (FAT)", "Master boot records (MBR)", "Virtual file layers (VFS)",
"Directory entries (dentries)" ],
"correctIndex" : 3
}, {
"question" : "What is the primary role of the Virtual File System (VFS) layer in a kernel?",
"answers" : [ "To provide an abstract uniform interface for applications to access diverse file
systems", "To simulate internal hard drive arrays inside local ram spaces", "To buffer incoming
network package flows before delivery to sockets", "To manage virtual memory swap spaces on
physical storage disks" ],
"correctIndex" : 0
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